Scheme of Valuation : QP Code-25901547

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

Fourth Semester

Discipline Specific Core Course (DSC) - MG4DSCSTA202 - PROBABILITY DISTRIBUTIONS AND STATISTICAL INFERENCE

Part A (Answer any 7 questions. Each question carries 2 marks)

1. Any one application of type II beta distribution — 2 Marks
2. Becomes geometric distribution OR pdf of NB distribution with $r = 1$ — 2 Marks
3. $f(2) = \frac{6C_2 14C_4}{20C_6}$=0.38738 or substitution of values in hypergeometric pdf — 2 Marks
4. Variance $\mu_2 = \frac{pq}{(p+q)^2(p+q+1)}$ — 2 Marks
5. One limitation of BLLN OR statement of BLLN — 2 Marks
6. Answer — 2 Marks
7. Explanation of assigning ranks in one sample case — 2 Marks
8. F statistic formula — 2 Marks
9. Use of normal distribution in large sample test — 2 Marks
10. Any one difference OR definitions of parametric and non-parametric test —2 Marks.

Part B (Answer any 4 questions. Each question carries 6 marks)

11. Definition of log normal distribution with pdf — 3 Marks
    relation with normal distribution —3 Marks
12. Mean= E(X) –1 mark, Derivation of E(X) –2 Marks
    mgf $M_X(t)$=$E(e^{tX})$–1 Mark, Derivation of $M_X(t)$ –2 Marks
    OR
    Derivation of mgf and obtaining mean from mgf – 6 Marks

13. Identify the distribution :NB with x=3, r=4, p=0.8– 1 Mark
    pdf – 1 Mark
    P(3 failures before 4th success)= P(X=3)=0.065536 –2 Maks
    Expected Number= $E(X) = \frac{rq}{p}$=1 – 2 Marks

14. Tchebycheff's inequality for lower bound of probability – 2 Marks
    $P(30 < X < 70) = P(|X - 50| < 20)$– 1 Mark
    t =4 – 2 Marks, lower bound =0.9375– 1 Mark

15. Explanation of sign test for one sample or two samples — 4 Marks
    Any four assumptions — 0.5 Marks each

16. Hypothesis – 1 mark, Test statistic — 1 Mark, Critical region – 1 Mark
    Calculated value of Z = -1.74078 –1 mark, table value $Z_{\frac{\alpha}{2}}$ =1.96 – 1 Mark
    Conclusion – 1 Mark

Part C (Answer any 1 Question. Each question carries 12 Marks)

17. Pdf or properties of binomial distribution – 4 Marks
    pdf or properties of Poisson distribution – 4 Marks
    pdf or properties of normal distribution –4 Marks
    OR
    any reasonable attempt stating binomial/ Poisson tend to normal by CLT – 12 Marks

18. Null and alternative hypothesis - 1 Mark
    Ranks of the combined sample – 2 Marks

Sample	I	I	I	I	II	I	II	II	I	II	II	II
Data	2	3	4	5	5	6	6	7	8	8	9	10
Rank	1	2	3	4.5	4.5	6.5	6.5	8	9.5	9.5	11	12

    Sum of ranks in Batch I = 26.5 – 1 Mark
    Sum of ranks in Batch II = 51.5 – 1 Mark
    U_1= 30.5 – 1.5 marks, U_2=5.5 –1.5 Marks Test statistic, U =$min(U_1, U_2)$ – 2 Marks
    Decision — 1 mark, Conclusion – 1 Mark